



- What is the value of $\log_{x \rightarrow \infty} (x + 1) \left(E^{\frac{1}{x+1}} - 1 \right)$?
(a) Does not exist (b) 0
(c) -1 (d) 1
- Number of three digit numbers that can be formed using 0, 1, 2, 3 and 5 where these digits are allowed to repeat any number of times, is equal to
(a) 125 (b) 100
(c) 115 (d) 140
- The circle $x^2 + y^2 + \alpha x + \beta y + \gamma = 0$ is the image of the circle $x^2 + y^2 - 6x - 10y + 30 = 0$ across the line $3x + y = 2$. The value of $[\alpha + \beta + \gamma]$ is (where $[\cdot]$ represents the floor function)
(a) 23 (b) 22
(c) 20 (d) 21
- Number of permutations of the letters of the word BANGLORE such that the string ANGLE appears together in all permutations, is
(a) 720 (b) 24
(c) 1228 (d) 120
- If $\vec{a}, \vec{b}$ and $\vec{c}$ are three vectors such that $\vec{a} \times \vec{b} = \vec{c}, \vec{a}, \vec{c} = 2$ and $\vec{b} \cdot \vec{c} = 1$. If $|\vec{b}| = 1$ then the value of $|\vec{a}|$ is
(a) 2 (b) 1
(c) 4 (d) 3
- The value of $\frac{d}{dx} \int_{\sin x}^{2\sin x} e^{t^2} dt$ at $t = \pi$ is
(a) -1 (b) 1
(c) -2 (d) 2
- The captains of five cricket teams, including India and Australia, are lined up randomly next to one other for a group photo. What is the probability that the captains of India and Australia will stand next to each other?
(a) $\frac{1}{4}$ (b) $\frac{2}{5}$
(c) $\frac{1}{2}$ (d) $\frac{1}{5}$
- Let $g : \mathbb{R} \rightarrow \mathbb{R}$ & $h : \mathbb{R} \rightarrow \mathbb{R}$, be two functions such that $h(x) = s g n(g(x))$. Then select which of the following is **not** true? ($\mathbb{R}$ denotes the set of all real numbers, $s g n$ stands for signum function)
(a) The domain of $h(x)$ is different from the domain of $g(x)$ at the same point.
(b) $h(x)$ is discontinuous at $g(x) = 0$.
(c) The domain of $h(x)$ is the same as the domain of $g(x)$.
(d) The domain of continuity of $h(x)$ equal the domain of continuity of $g(x) - \{x \in \mathbb{R}, g(x) = 0\}$
- What is the general solution of the equation $\cot \theta + \tan \theta = 2$?
(a) $\theta = \frac{n\pi}{2} + \frac{\pi}{4}$ (b) $\theta = \frac{n\pi}{2} + \frac{\pi}{6}$
(c) $\theta = n\pi + \frac{\pi}{4}$ (d) $\theta = \frac{n\pi}{2} + \frac{\pi}{8}$
- The obtuse angle between lines $2y = x + 1$ and $y = 3x + 2$ is
(a) $\frac{3\pi}{4}$ (b) $\frac{5\pi}{6}$ (c) $\frac{4\pi}{3}$ (d) $\frac{2\pi}{3}$
- The length of the projection of $\vec{a} = 2\hat{i} + 3\hat{j} + \hat{k}$ on $\vec{b} = -2\hat{i} + \hat{j} + 2\hat{k}$, is equal to
(a) $\frac{2}{3}$ (b) $\frac{-1}{3}$ (c) $\frac{-2}{3}$ (d) $\frac{1}{3}$
- The area enclosed between the curve $y = \sin x, y = \cos x, 0 \leq x \leq \frac{\pi}{2}$ is
(a) $(\sqrt{2} - 1)$ (b) $2(\sqrt{2} + 1)$
(c) $(\sqrt{2} + 1)$ (d) $2(\sqrt{2} - 1)$
- A circle with its center in the first quadrant touches both the coordinate axes and the line $x - y - 2 = 0$. Then the area of the circle is
(a) 2π (b) π (c) $\frac{\pi}{2}$ (d) 4π
- Let $\vec{a} = e\hat{i} - 3\hat{j} + 4\hat{k}, \vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j} + \lambda\hat{k}$ be the co-terminal edges of a parallelopioped whose volume is 5 units. Then the value of λ is
(a) 2 (b) 1 (c) 4 (d) 3
- If $8^{x-1} = \left(\frac{1}{4}\right)^x$, then the value of $\frac{1}{\log_{x+1} 4 - \log_{x+1} 5} + \frac{1}{\log_{1-x} 4 - \log_{1-x} 5}$ is
(a) $\frac{5}{4}$ (b) 1 (c) 2 (d) $\frac{4}{5}$
- If α and β are the two roots of the equation $x^2 + ax + b = 0, (ab \neq 0)$ then the quadratic equation with roots $\frac{1}{\alpha^3 + \alpha}$ and $\frac{1}{\beta^3 + \beta}$ is
(a) $b(b^2 + 1 + a^2 - 2b)x^2 + (a^3 + a - 3ab)x + 1 = 0$
(b) $b(b^2 + 1 + a^2 + 2b)x^2 + (a^3 - a - 3ab)x + 1 = 0$
(c) $b(b^2 + 1 + a^2 + 2b)x^2 - (a^3 + a - 3ab)x + 1 = 0$
(d) $b(b^2 + 1 + a^2 - 2b)x^2 - (a^3 + a - 3ab)x + 1 = 0$
- Let E and F be two events such that $P(E) > 0$ and $P(F) > 0$. Which one of the following is **NOT** equivalent to the condition that $P(E) = P(EIF)$?
(a) $P(F) = P(FE)$
(b) E and F are independent
(c) E^c and F are independent
(d) $P(E^c) P(F^c) \neq P(E^c \cap F^c)$
- Let the line $\frac{x}{4} + \frac{y}{2} = 1$ meets the x-axis and y-axis at A and B, respectively. M is the midpoint of side AB, and M' is the image of the point M across the line $x + y = 1$. Let the point P lie on the line $x + y = 1$ such that the ΔABP is an isosceles triangle with $AP = BP$. Then the distance between M' and P is
(a) $\frac{\sqrt{7}}{3}$ (b) $\frac{2\sqrt{5}}{3}$ (c) $\frac{2\sqrt{7}}{3}$ (d) $\frac{\sqrt{5}}{3}$



19. Let $\vec{a}$ and $\vec{c}$ be unit vectors such that the angle between them is $\cos^{-1}\left(\frac{1}{4}\right)$. If $\vec{b} = 2\vec{c} + \lambda\vec{a}$. Where $\lambda > 0$ and $|\vec{b}| = 4$, then λ is equal to
(a) 4 (b) 3 (c) $\frac{1}{3}$ (d) $\frac{1}{4}$
20. A tower subtends angles α , 2α and 3α , respectively at points A, B and C which are lying on a horizontal line through the foot of the tower. Then $\frac{AB}{BC}$ is equal to
(a) $1 + 2\cos 2\alpha$ (b) None of these
(c) $2 + \cos 3\alpha$ (d) $2 + 3\cos 3\alpha$
21. If x , y and z are the three cube roots of 27, then the determinant of the matrix $\begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}$ is
(a) 0 (b) -1
(c) 1 (d) $(x-y)(x-z)(y-z)$
22. There are two coins, say blue and red. For blue coin, probability of getting head is 0.99 and for red coin, it is 0.01. One coin is chosen randomly and is tossed. The probability of getting head is
(a) 1 (b) 0.98 (c) 0.5 (d) 0.02
23. Let $A = \{1, 2, 3, \dots, 20\}$. Let $R \subseteq A \times A$ such that $R = \{(x, y) : y = 2x - 7\}$. Then the number of elements in R , is equal to
(a) 7 (b) 13 (c) 17 (d) 10
24. Let A and B be two square matrices of same order satisfying $A^2 + 5A + 5I = 0$ and $B^2 + 3B + I = 0$ respectively. Where I is the identity matrix. Then the inverse of the matrix $C = BA + 2B - 2A + 4I$
(a) $AB + A + 3B + 3I$ (b) $BA + 3B + A + 3I$
(c) $BA - 3B + A - 3I$ (d) $AB - A + 3B - 3I$
25. Consider the sample space $\Omega = \{(x, y) : x, y \in \{1, 2, 3, 4\}\}$ where each outcome is equally likely. Let $A = \{x \geq 2\}$ and $B = \{y > x\}$ be two events. Then which of the following is **NOT** true?
(a) $A \cap B = 1/4$
(b) A and B are not independent
(c) $P(B) = 3/8$
(d) $P(A) = 3/4$
26. The number of all even integers between 99 and 999 which are not multiple of 3 and 5 is
(a) 240 (b) 250 (c) 235 (d) 245
27. The curve $y = \frac{x}{1 + x \tan x}$ attains maxima
(a) at $x = \cos x$
(b) at $x = -\cos x$
(c) at $x = \cos x$ and for some $x \in [-2\pi, \pi]$
(d) for some $x \in [-2\pi, \pi]$
28. There are 40 female and 20 male students in a class. If the average heights of female and male students are 5.15 feet and 5.66 feet, respectively, then the average height (in feet) of all the students in the class equals
(a) 5.490 (b) 5.405 (c) 5.175 (d) 5.320
29. The maximum value of $\sin(x) + \sin(x + 1)$ is $k \cos \frac{1}{2}$. Then the value of k is
(a) 1 (b) 2
(c) None of these (d) 3
30. Let $A = \{5^n - 4n - 1 : n \in \mathbb{N}\}$ and $B = \{16(n - 1) : n \in \mathbb{N}\}$ be sets. Then
(a) Neither $A \subset B$ nor $B \subset A$
(b) $A \subset B$
(c) $B \subset A$
(d) $A \cap B$ is a finite set
31. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, be any function defined as $f(x) = \begin{cases} x^\alpha \sin \frac{1}{x^\beta}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, $\alpha, \beta \in \mathbb{R}$, which of the following is true? ($\mathbb{R}$ denotes the set of all real numbers)
(a) $f(x)$ is continuous at $x = 0$, for all α & $\beta \in \mathbb{R}$
(b) $f(x)$ is differentiable at $x = 0$ for all $\alpha > 0$ & $\beta > 0$
(c) $f(x)$ is continuous at $x = 0$ for only $\alpha > 0$ & $\beta > 0$
(d) $f(x)$ is continuous at $x = 0$ for all $\alpha > 0$ & $\beta \in \mathbb{R}$
32. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{c} = \hat{i} - 2\hat{j} + \hat{k}$, then a vector of magnitude $\sqrt{22}$ which is parallel to $2\vec{a} - \vec{b} + 3\vec{c}$ is
(a) $3\hat{i} + 3\hat{j} + 2\hat{k}$ (b) $(\hat{i} - 4\hat{j} + 2\hat{k})\sqrt{\frac{22}{21}}$
(c) $3\hat{i} - 3\hat{j} - 2\hat{k}$ (d) $(\hat{i} - 4\hat{j} - 2\hat{k})\sqrt{\frac{22}{21}}$
33. An equilateral triangle is inscribed in the parabola $y^2 = x$. One vertex of the triangle is at the vertex of the parabola. The centroid of triangle is
(a) $(2, 0)$ (b) $(\sqrt{2}, 0)$
(c) $(1, 0)$ (d) $(\sqrt{3}, 0)$
34. A tower subtends an angle of 30° at a point on the same level as the foot of the tower. At a second point h meters above the first, the depression of the foot of the tower is 60° . What is the horizontal distance of the tower from the point?
(a) $2h \cot 60^\circ$ (b) $2h \cot 30^\circ$
(c) $h \tan 60^\circ$ (d) $2h \tan 60^\circ$
35. The angles of depression of the top and bottom of an 8m tall building from the top of a multi-storied building are 30° and 45° , respectively. What is the height of the multi-storied building and the distance between the two buildings?
(a) $4(3 + \sqrt{3})$ m, $4(3 + \sqrt{3})$ m
(b) $2(3 + \sqrt{3})$ m, $2(3 + \sqrt{3})$ m
(c) $4(2 + \sqrt{2})$ m, $4(2 + \sqrt{2})$ m
(d) $2(3 - \sqrt{3})$ m, $4(3 + \sqrt{3})$ m
36. If $\cos^2(10^\circ) \cos(20^\circ) \cos(40^\circ) \cos(50^\circ) \cos(70^\circ) = \alpha + \frac{\sqrt{3}}{16} \cos(10^\circ)$, then $3\alpha^{-1}$ is equal to
(a) $\frac{9}{32}$ (b) 64 (c) 16 (d) $\frac{9}{16}$





37. Suppose $t_1, t_2, t_3, \dots, t_{55}$ are in AP such that $\sum_{l=0}^{18} t_{3l+1} = 1197$ and $t_7 + 3t_{22} = 174$. If $\sum_{t=1}^9 t_l^2 = 947b$, then the value of b is
(a) 5 (b) 3 (c) 1 (d) 2
38. Consider the matrix $B = \begin{pmatrix} -1 & -1 & 2 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{pmatrix}$. The sum of all the entries of the matrix B^{19} is
(a) -171 (b) -192 (c) -174 (d) -163
39. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 3, |\vec{b}| = 4$ and $|\vec{a} + \vec{b}| = 1$, then the value of $|\vec{a} - \vec{b}|$ is
(a) 7 (b) 2 (c) 6 (d) 1
40. The scores of students in a national level examination are normally distributed with a mean of 500 and a standard deviation of 100. If the value of the cumulative distribution of the standard normal random variable at 0.5 is 0.691, then the probability that a randomly selected student scored between 450 and 500 is
(a) 0.191 (b) 0.391 (c) 0.591 (d) 0.091
41. Which one of the following is **NOT** a correct statement?
(a) The standard deviation is greater than or equal to the mean deviation (about mean)
(b) The variance is expressed in the same units as the units of observation
(c) The value of standard deviation changes by a change of scale.
(d) The sum of squares of deviations is minimum when taken from the mean
42. Let F_1, F_2 be the foci of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, a > 0, b > 0$, and let O be the origin. Let M be an arbitrary point on curve C and above X-axis and H be a point on MF_1 such that $MF_2 \perp F_1 F_2, MF_1 \perp OH, |OH| = \lambda |OF_2|$ with $\lambda \in (2/5, 3/5)$, then the range of the eccentricity e in
(a) $(1, \sqrt{7/3})$ (b) $(\sqrt{7/3}, 2)$
(c) $(\sqrt{2}, \sqrt{3})$ (d) $(\sqrt{3}, 2)$
43. The value of the integral $\int_0^{\frac{\pi}{2}} \frac{(1 + 2 \cos x)}{(2 + \cos x)^2} dx$ lies in the interval
(a) (-2, -1) (b) (-1, 0)
(c) (0, 1) (d) (1, 2)
44. Let x be a positive real number such that $x^{(8 \log_5(x) - 24)} = 5^{-4}$. Then the product of all possible values of x is
(a) 125 (b) 625 (c) 25 (d) 5

45. Given the equation $x + y = 1, x^2 + y^2 = 2, x^5 + y^5 = A$. Let N be the number of solution pairs (x, y) to this system of equations. Then AN is equal to
(a) 9 (b) 3
(c) $\frac{7}{2}$ (d) $\frac{19}{2}$
46. The number of accidents per week in a town follows Poisson distribution with mean 2. If the probability that there are three accidents in two weeks time is ke^{-6} , then the value of k is
(a) 18 (b) 36 (c) 27 (d) 9
47. The slope of the normal line to the curve $x = t^2 + 3t - 8$ and $y = 2t^2 - 2t - 5$ at the point $(2, -1)$ is
(a) $\frac{22}{7}$ (b) -5
(c) $-\frac{6}{7}$ (d) $-\frac{7}{6}$
48. If $B = \sin^2 y + \cos^4 y$, then for all real y
(a) $\frac{3}{4} \leq B \leq 1$ (b) $\frac{3}{4} \leq B \leq \frac{13}{16}$
(c) $\frac{13}{16} \leq B \leq 1$ (d) $1 \leq B \leq 2$
49. The number of 3-digit integers that are multiple of 6 which can be formed by using the digits 1, 2, 3, 4, 5, 6 without repetition is
(a) 22 (b) 26 (c) 20 (d) 24
50. An airplane, when 4000m high from the ground, passes vertically above another airplane at an instant when the angles of elevation of the two airplanes from the same point on the ground are 60° and 30°, respectively. Find the vertical distance between the two airplanes
(a) 16000/7m (b) 8000/3m (c) 8000/7m (d) 1200m

REASONING

1. Statements:

- I. All cakes are ice cream
II. No ice cream is chocolate

Conclusions:

- I. Some cakes are chocolate
II. No cake is chocolate

Choose the correct option:

- (a) Neither conclusion I nor conclusion II follows
(b) Only conclusion I follows
(c) Both conclusion I and conclusion II follows
(d) Only conclusion II follows





2. A, B, C, D, E, F and G are travelling in three different vehicles: Swift, Creta, and Nexon.
There are at least two passengers in each vehicle. Among them, only two are male. There are two engineers, two doctors and three teachers
(i) C is a lady doctor and she does not travel with A and F, who are sisters.
(ii) B, a male engineer, travels with only G, a teacher, in a Swift.
(iii) D is a male doctor
(iv) Two persons belonging to the same profession do not travel in the same vehicle.
(v) A is not an engineer and travels in a Creta.
(vi) The pair of sisters A and F travels in the same vehicle
What is the profession of F?
(a) Engineer (b) Teacher
(c) Data Inadequate (d) Doctor
3. If 1 man or 2 women can finish a job in 20 days, how many days will 3 men and women take to finish the same job?
(a) 3 (b) 5 (c) 6 (d) 4
4. Consider a sequence formed by concatenating the positive integers in increasing order: 12345678910111213..... Which among the following is the closest to the 2028th digit in this infinite sequence?
(a) 4 (b) 7 (c) 1 (d) 9
5. **Statement:** All engineers are good at math.
Conclusion 1: Some people good at math are engineers.
Conclusion 2: All people good at math are engineers.
(a) Only Conclusion 1 follows
(b) Only Conclusion 2 follows
(c) Both Conclusions follow
(d) Neither follows
6. A number which, when increased by 15% becomes 161. What is the original number?
(a) 135 (b) 140 (c) 145 (d) 130
7. Two ships are sailing in the sea on the two sides of a lighthouse. The angle of elevation of the top of the lighthouse is observed from the ships are 30° degree and 45° respectively. If the lighthouse is 100 m high, the distance between the two ships is
(a) 300 m (b) 200 m
(c) 173 m (d) 273 m
8. Mohan introduced Sarita as the mother of his sister's brother. How is Sarita related to Mohan?
(a) Mother (b) Niece
(c) Sister (d) Aunt
9. Ten years ago, Ram was twice as old as Shyam. Ten years from now, the ratio of their ages will be 3:2. What is Ram's present age?
(a) 50 (b) 60 (c) 40 (d) 70
10. "In order to be a teacher one must graduate from college. All poets are poor. Some Mathematicians are poets. No college graduate is poor."
(a) Some Mathematicians are not teachers.
(b) Poets are not teachers.
(c) Teachers are not poor.
(d) Some teachers are not Mathematicians.
11. Consider the following statements followed by two conclusions.
Statements: 1. Some men are great. 2. Some men are wise
Conclusions: 1. Men are either great or wise. 2. Some men are neither great nor wise.
(a) Both the conclusions are valid
(b) Only conclusion 2 is valid
(c) Only conclusion 1 is valid
(d) Neither of the conclusions is valid.
12. If John's mother's father's granddaughter is Maria, what is Maria's relation to John?
(a) Sister (b) Niece
(c) Aunt (d) Cousin
13. If a number is increased by 30% and then decreased by 20%, what is the net change?
(a) 4% increase (b) 6% increase
(c) 8% Increase (d) 6% decrease
14. If 1st January 2022 was a Saturday, what day of the week will 1st January 2023 be?
(a) Monday (b) Tuesday
(c) Wednesday (d) Sunday
15. A person walks 8 km north, turns right, walks 4 km, turns right again and walks 6 km. Finally, he turns left and walks 3 km. How far is he from the starting point?
(a) 8.92 km (b) 5.89 km
(c) 7.62 km (d) 6.45 km
16. A two digit number is such that the product of its digits is 12. If 36 is added to the number, the digits interchange their places. What is the number?
(a) 24 (b) 38 (c) 26 (d) 25
17. In a remote research outpost, five scientists (Dr. Anand, Dr. Beena, Dr. Chirag, Dr. Dinesh, and Dr. Elango) are studying a rare phenomenon. They communicate through a secure, text-based system. Over the course of an hour, the following exchanges occur:
1. Dr. Anand sends a message to Dr. Beena and Dr. Chirag
2. Dr. Beena replies only to Dr. Anand.
3. Dr. Anand sends a reminder message to who did not reply.
4. Dr. Chirag sends a message to Dr. Dinesh and Dr. Elango.
5. While replying to Dr. Chirag, Dr. Dinesh accidentally sends a "Reply All"
6. Dr. Elango, having received Dr. Chirag's message, sends a message to Dr. Beena.





At the end of this hour, who has received the most messages directly from other scientists within the group? (Consider only messages sent within the described hour and only between the five scientists mentioned)

- (a) Dr. Chirag (b) Dr. Dinesh
(c) Dr. Anand (d) Dr. Beena

18. A remote island has a unique social structure. Individuals are either 'Truth-tellers' (who always speak the truth) or 'Tricksters' (who always lie). You encounter three inhabitants: X, Y, and Z.

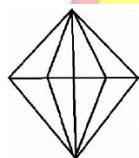
X says: "Y is a Trickster"

Y says: "Exactly one of us is a Truth-teller."

What can you definitively conclude about 27

- (a) Z could be either a Truth-teller or a Trickster
(b) Z is a Trickster
(c) Z is a Truth-teller
(d) The information is insufficient to determine Z's identity

19. How many triangles are there in the figure given below?



- (a) 13 (b) 21 (c) 24 (d) 32

20. What is the next number in each of the following 3 sequences?

8 17 33 67 133? (1)

7 23 67 203 607? (2)

6 27 105 423 1689? (3)

(a) 267, 1823, 6759 (b) 213, 1534, 7635

(c) 238, 1846, 6389 (d) 254, 968, 3689

21. In a certain code language, the word "REASON" is encoded as TDDQSK. Using the same rule, how is the word PRAYER coded?

(a) RTBXCT (b) RQDWIO

(c) SSAYDU (d) RQDZCU

22. A monkey climbs 30 feet at the beginning of each hour and rests for a while when he slips back 20 feet before he again starts climbing in the beginning of the next hour. If he begins his ascent at 8.00 am, at what time will he first touch a flag at 120 feet from the ground!

(a) 5 p.m. (b) 9 p.m

(c) 6 p.m (d) 8 p.m

23. At an IIM entrepreneurship summit, two young founders — Karan and Deepak — introduced their startup. In their quirky opening, they said "The product of our ages is 240. And just like in our startup strategy, twice Deepak's age is 4 years more than Karan's age. How old was Deepak two years ago?

(a) 8 Years (b) 15 Years

(c) 6 Years (d) 10 Years

24. Four people (P, Q, R & S) are standing in a queue. Q is ahead of P. R is between P and S. If S is at the end of the queue, who is at the front of the queue?

(a) Q (b) R

(c) P (d) S

25. A research team is studying the effects of three different fertilizers (X, Y, and Z) on the growth of a specific plant species. They have four experimental plots with different fertilizer combinations

- Plot 1: Fertilizer X only
- Plot 2: Fertilizer Y only
- Plot 3: Fertilizer X and Fertilizer Z only
- Plot 4: Fertilizer Y and Fertilizer Z only

After a period of observation, they note the following

- Plants in plots with Fertilizer X showed significantly increased height compared to a control group (no fertilizer)
- Plants in plots with Fertilizer Y showed a slightly increased leaf area compared to the control group.
- Plants in Plot 3 (X and Z) showed no significant difference in height compared to the control group.
- Plants in Plot 4 (Y and Z) showed significantly increased height compared to the control group

Based on these observations, which of the following conclusions is best supported?

(a) Fertilizer Z inhibits the height growth promoted by Fertilizer X.

(b) Fertilizer Z enhances the leaf area increase caused by Fertilizer Y.

(c) Fertilizer X is the only fertilizer that significantly increases the height of this plant species.

(d) Fertilizer Z has no effect on the growth of this plant species.

26. Which number replaces the question mark?

4, 9, 16, 25, ?, 49

(a) 40 (b) 36

(c) 32 (d) 30

27. In a box, there are 8 red, 7 blue and 6 green balls. One ball is picked up randomly. What is the probability that it is neither red nor green?

(a) $\frac{3}{4}$ (b) $\frac{7}{19}$

(c) $\frac{1}{3}$ (d) $\frac{8}{21}$

28. Which number replaces the question mark?

3, 6, 18, 72, ?, 432

(a) 360 (b) 144

(c) 288 (d) 216

29. In a certain code language, 'SUN' is written as 'RWK' and 'MOON' is written as 'LQLK'. Following the same pattern, how will 'STAR' be written?

(a) RUDY (b) RUXV

(c) SVXU (d) RVXV





30. Five friends A,B,C,D and E are sitting in a row. The following conditions hold:
A is sitting next to B.
C is sitting at one of the ends.
D is sitting between A and E
E is sitting to the right of D.
Who is sitting at the second position?
(a) E (b) A (c) D (d) C
31. **Statements:**
All apples are fruits. All fruits are tasty.
Conclusions:
I. All apples are tasty.
II. Some tasty things are apples.
(a) Neither follows (b) Only I follows
(c) Both follow (d) Only II follows
32. Average Cost (AC) is defined as the sum of Average Fixed Cost (AFC) and Average Variable Cost (AVC), and dumping is defined as selling a product at a price less than AC but more than AVC. A company in India, suddenly, found that the demand for its product "ZOOM" has fallen to 60% of the output produced in that financial year. As a result, the company must sell 40% of the produced output in a foreign market. If it decides to "dump" 40% of its output in a neighbouring country (by reducing the price by 20%), what would be the objective of its 'dumping strategy', among the following? (The company has set a profit margin of 10% of AC while fixing the price of its product for sale in India).
(A) To minimize losses,
(B) to maximize profits,
(C) to contribute to the recovery of fixed costs,
(D) to contribute to the recovery of variable costs
(a) (A) and (B) (b) (A) and (C)
(c) (B) and (D) (d) (B) and (C)
33. A group of 630 children is arranged in rows for a group photograph. Each row contains three fewer children than the row in front of it. What number of rows is not possible?
(a) 6 (b) 3
(c) 5 (d) 4
34. Complete the next series:
2 8 12
8 40 45
40 240 246
--- --- ---
(a) 160, 960, 967 (b) 240, 1680, 1687
(c) 200, 1000, 1005 (d) 48, 288, 295
35. The percentage change in area of a rectangle by decreasing its length and increasing its breadth by 5% is
(a) no change (b) 0.25% decrease
(c) 2.5% increase (d) 0.25% increase

36. **Statements:**
I. The meteorological department predicts deficient rainfall during monsoon season
II. The GDP growth estimates have been revised downwards by the government.
Choose the correct option
(a) Statement I is the cause and Statement II is the effect
(b) Both the statements I and II are effects of independent causes
(c) Both the statements I and II are independent causes
(d) Both the statements I and II are effects of some common cause
37. Read definition and all four choices carefully, and find the answer that provides the best example of the given definition. Answer the question solely on the basis of the definition given
Definition:
An informal gathering occurs when a group of people get together in a casual, relaxed manner.
Which situation below is the best example of an informal gathering?
(a) Mary sends out 25 invitations for the bridal shower she is giving for her sister
(b) The book club meets on the first Thursday evening of every month
(c) Whenever she eats at the Italian restaurant, Kamalika seems to run into Peter
(d) After finding about his promotion, Jeremy and a few coworkers decide to go out for a quick drink after work
38. A sum of money was divided into 2 parts in the ratio 2:5. First part was invested for 2 years at the annual interest rate of 20% compounded annually. At what rate of simple interest per annum the second part must be invested for 2 years, so that the interest earned in both cases is the same?
(a) 9.6%
(b) 8.8%
(c) Cannot be determined
(d) 10.2%
39. What comes next in the sequence?
2, 6, 12, 20, 30, _____
(a) 36 (b) 50
(c) 56 (d) 42
40. Which word in the following is the "odd man" out?
(a) Pampering
(b) Promoting
(c) Nurturing
(d) Incubating





COMPUTER

- Dynamic RAM (DRAM) stores each bit of data in a separate capacitor. Due to leakage, the stored charge tends to dissipate over time and needs to be refreshed periodically. Consider the following statements:
P: DRAM requires refreshing because it uses capacitors to store bits.
Q: SRAM does not require refreshing because it uses flip-flops instead of capacitors.
R: DRAM is faster than SRAM because it needs less frequent access.
S: DRAM is more suitable for main memory than SRAM due to its density.
(a) All statements P, Q, R, and S are true
(b) P, Q, and S are true; R is false.
(c) Only P and Q are true; R is false, and S is irrelevant.
(d) P and S are true; Q and R are false.
- Suppose that C represents the set of all countries, R represents the set of all countries that have at least one river flowing through it, M represents the set of all countries that have at least one mountain in it, and D represents the set of all countries that have at least one desert in it. It is given that $(R \cup M \cup D) = C$. Which one of the following gives the set of all countries that have either a mountain or a river, but does not have a desert in it? The notation D^c represents the complement of the set D with respect to the universal set C.
(a) $(R \cup M) - (R \cap M \cap D^c)$
(b) $(R \cap M) \cap D^c$
(c) $(R \cup M) \cap D^c$
(d) $(R \cup M) - (R \cap M)$
- Consider a system with a CPU having 6 registers and 32-bit instructions. The maximum possible size of the main memory is 512 KB ($1K = 2^{10}$). Each instruction takes two registers and one memory address as operands. Which one of the following correctly gives the maximum possible distinct instructions that can be there in the instruction set of the CPU?
(a) 128 (b) 256 (c) 64 (d) 1024
- Which of the following statements about ASCII and Unicode is correct?
(a) ASCII uses 16 bits per character, while Unicode uses only 7 bits.
(b) Unicode is backward compatible with ASCII and includes all ASCII characters in its encoding.
(c) Unicode and ASCII are completely different and share no common characters.
(d) ASCII can represent more characters than Unicode because it uses fewer bits per character.
- In computer architecture, which of the following correctly matches the fundamental unit used in the CPU, cache, and main memory respectively?
(a) CPU → Block, Cache → Word, Memory → Page
(b) CPU → Word, Cache → Block, Memory → Page
(c) CPU → Page, Cache → Word, Memory → Block
(d) CPU → Block, Cache → Page, Memory → Word
- Debugger is a program that
(a) does not allow step by step execution of a segment of program
(b) compiles source code to object code
(c) links object code to produce an executable
(d) allows to set breakpoints, execute a segment of program and display contents of Register
- Consider a system running under two types of workloads: (a) CPU-intensive jobs, (b) I/O-intensive jobs. Which of the following statements about the relative performance of Interrupt-driven I/O and Programmed I/O is correct?
(a) Programmed I/O performs better for CPU-intensive jobs, while interrupt-driven I/O performs better for I/O-intensive jobs
(b) Interrupt-driven I/O performs better in both CPU-intensive and I/O-intensive workloads because it always reduces CPU involvement.
(c) Interrupt-driven I/O performs better for CPU-intensive jobs by freeing the CPU, while programmed I/O performs better for I/O-intensive jobs due to tighter control.
(d) Programmed I/O performs better in both CPU-intensive and I/O-intensive workloads because it gives the CPU full control.
- What is the equivalent decimal value of the unsigned binary number 10101.10101?
(a) 51.125 (b) 125.25
(c) 25.375 (d) 21.65625
- Consider the Boolean expression $X = \overline{(A + B)} \cdot C$ over two Boolean variables A and B. Which one of the following Boolean expressions is equivalent to the given Boolean expression X?
(a) $\overline{A} \cdot \overline{B} + C$ (b) $\overline{A} \cdot \overline{C} + \overline{B} \cdot \overline{C}$
(c) $\overline{A} \cdot \overline{B} + \overline{C}$ (d) $\overline{A} + \overline{B} \cdot \overline{C}$
- Given the Boolean expression $(A \oplus B) \wedge (B \rightarrow C)$, which of the following rows in the truth table would have an output of 1 (True)?
(a) $A = 1, B = 1, C = 1$ (b) $A = 0, B = 1, C = 0$
(c) $A = 0, B = 0, C = 0$ (d) $A = 1, B = 0, C = 1$
- Consider the Boolean function in a CPU control unit:
 $Y = (A + B \cdot \overline{C}) \cdot (\overline{A} \cdot B + C)$
(A) $A \cdot B + C$ (B) $A \cdot C + \overline{A} \cdot B + B \cdot \overline{C}$
(C) $A \cdot C + \overline{A} \cdot B \cdot \overline{C}$ (D) $A \cdot C + B \cdot \overline{C}$





12. Given an unsigned 32-bit integer x , which of the following C / C++ expressions correctly toggles m bits starting from position p (with the least significant bit at position 0)?

Assume: x is the input integer; p is the starting position of the bit range (0-based, LSB at position 0); m is the number of bits to toggle; and No overflow or invalid input conditions occur.

Which of the following correctly toggles m bits of x starting from position p ?

- (a) $x^{\wedge}(((1 \ll m) - 1) \ll p)$
(b) $x|((1 \ll m) \ll p)$
(c) $x^{\wedge}((1 \ll m) \ll p)$
(d) $x\&\sim(((1 \ll m) - 1) \ll p)$

13. In an instruction execution pipeline, the earliest that the instruction TLB and data TLB can be accessed are

- (a) Fetch stage and fetch stage respectively
(b) Fetch stage and memory stage respectively
(c) Memory stage and execute stage respectively
(d) Memory stage and memory stage respectively

14. Which of the following secondary storage devices has the fastest access time: Optical Drive, Magnetic Tape Drive, Hard Disk Drive (HDD), Solid State Drive (SSD)?

- (a) Solid State Drive (SSD) (b) Optical Drive
(c) Magnetic Tape Drive (d) Hard Disk Drive (HDD)

15. What is the hexadecimal representation of the decimal number 265?

- (a) 0x100001001 (b) 0x411
(c) 0x190 (d) 0x109

16. Consider a 9-bit representation. Which of the following correctly gives the smallest number that can be represented in: (i) 1's complement, (ii) 2's complement

- (a) (i) -256, (ii) -255 (b) (i) -255, (ii) -255
(c) (i) -256, (ii) -256 (d) (i) -255, (ii) -256

17. In the design of a control unit of a processor, two common approaches are used: hardware control and microprogrammed control. Consider the following statements:

- I. Hardware control units are generally faster but more difficult to modify than microprogrammed control units.
II. In a horizontal microprogrammed control unit, each control signal has a separate bit in the control word.
III. Vertical microprogramming leads to longer control words but provides greater parallelism.
IV. Microprogrammed control units are typically easier to implement and modify than hardware control units

- (a) I, II, and IV only (b) I, II, III, and IV
(c) II, III, and IV only (d) I, III, and IV only

18. In 8-bit two's complement arithmetic, compute the result of the following addition: $A = 10011001$, $B = 11010111$. What is the resulting 8-bit binary value?

- (a) 01110000
(b) 11100100
(c) 01110001
(d) Result is invalid

19. Which of the following is the CORRECT truth table for the XOR Gate with two binary inputs A and B ?

(a)

A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

(b)

A	B	A XOR B
0	0	1
0	1	1
1	0	1
1	1	0

(c)

A	B	A XOR B
0	0	1
0	1	0
1	0	0
1	1	1

(d)

A	B	A XOR B
0	0	0
0	1	0
1	0	1
1	1	1

20. Match all items in Group 1 with correct options from those given in Group 2

Group 1	Group 2
P. Intermediate representation	1. Activation records
Q. Top-down parsing	2. Code generation
R. Runtime environments	3. Leftmost derivation
S. Register Allocation	4. Graph colouring

- (a) P-4, Q-2, R-3, S-1 (b) P-1, Q-2, R-3, S-4
(c) P-2, Q-3, R-1, S-4 (d) P-2, Q-3, R-4, S-1





ENGLISH

21. After running around the city all day delivering invitations for the wedding I desperately needed to hit the sack. (Identify the meaning of the underlined phrases)
 - (a) To pack one's bags
 - (b) To exercise for fitness
 - (c) To go to bed
 - (d) to watch television
22. Even though in official statements, the Presidential office has warned against violating the 25th protocol, the government's inaction against the Market group shows the government's tacit approval of the movement. (Choose the antonym of the underlined word)
 - (a) Undeclared
 - (b) Implicit
 - (c) Unequivocal
 - (d) Unvoiced
23. The new iPhone seems to cost everyone not just an arm and a leg, but also a kidney. "What does "to cost an arm and a leg" mean?
 - (a) to be very cheap
 - (b) To be very expensive
 - (c) To be afraid of surgery
 - (d) to be very short-lived
24. What is the synonym of "belligerent"?
 - (a) Equanimity
 - (b) Hostile
 - (c) Bellyful
 - (d) Peaceful
25. Which of the following sentences uses the correct form of the very "to lay"?
 - (a) She lays on the bed every night.
 - (b) She is laying down for a nap.
 - (c) She has lain the keys on the counter.
 - (d) She laid the book on the table.
26. Study the following sentence: The statue of Christ the Redeemer is located on Rio de Janeiro, the largest city of Brazil. (Which of the following parts of the sentence is grammatically incorrect)
 - (a) on Rio de Janeiro
 - (b) The statue of Christ the Redeemer
 - (c) is located
 - (d) the largest city of Brazil

27. Which of the following statements correctly explains the use of articles in the following sentence. After earning an MBA, Srinivas was able to approach a mathematical problem from both a business and analytical perspective.
 - (a) "An MBA" uses a because MBA is an acronym, and "a mathematical problem" should use the since it is referring to a unique or previously mentioned problem.
 - (b) "An MBA" should use a instead, because it refers to a specific degree, and "a mathematical problem" should use the to refer to a specific problem.
 - (c) "An MBA" uses the indefinite article an because it begins with a vowel sound, and "a mathematical problem" uses a because "mathematical" begins with a consonant sound.
 - (d) Both articles are used incorrectly because "an MBA" should use a and "a mathematical problem" should use the.
28. One of the employees _____ planning to transfer to another department.
 - (a) were
 - (b) are
 - (c) has
 - (d) is
29. The man and the dog _____ (flee) through the forest. (fill in the blank with the most appropriate form)
 - (a) has fled
 - (b) flees
 - (c) were fleeing
 - (d) fleeing
30. Identify the incorrect sentence based on the usage of the verb
 - (a) They are understanding the problem
 - (b) It's been raining since morning
 - (c) It has clear ideas
 - (d) Their understanding of the problem was incorrect

